Private Innovation Intermediation, Regional Incidence, and Public Access

Abstract

How do private innovation fees and local public-access decisions depend on regional commercialisation returns, service values and technical complementarity, and when do local and national provision evaluations diverge? We study two local governments and two private intermediaries with endogenous fees, founder selection, cross-region research matching and differentiated Cournot competition. Private pricing selects abilities and free basic access reduces private receipts without necessarily inducing exit. Conditional on full participation in the partner region, we derive separate local and national public-cost ceilings. A primitive profit-share condition shows that stronger technical complementarity can raise the local ceiling even when additional ventures reduce regional commercialisation surplus. This result is analytical for sufficiently weak matching; exact rational certification establishes its signs on a continuous two-parameter region at the original finite intensity. Three separately certified four-party entry examples compare decentralised provision with national welfare maximisation over sixteen configurations: two exhibit public underprovision and one coincidence. Their configuration switch requires a narrow public-cost band and fails a recorded alternative ability distribution. The paper provides conditional, model-specific policy boundaries, separating familiar screening and externality mechanisms from the candidate contribution of their regional-incidence interaction.

Keywords: Innovation intermediation; Public access; Participant selection; Mixed provision; Regional welfare.

JEL classification: D43; H77; L31; O32.

1 Introduction

Local governments provide free basic innovation intermediation while private specialists sell paid services to prospective founders. The welfare case for public access depends on more than the number of supported ventures. Regions can differ in the population seeking services, the competitive effects of commercialisation on existing producers and consumers, and the share of new venture profits retained by local founders. Technical complementarity between research partners can also change the value of access. A common service policy can consequently face different regional economic boundaries, even without differences in the underlying distribution of founder abilities.

Several constituent mechanisms are already understood. Private matching prices select heterogeneous participants and can affect match quality as well as volume (Damiano and Li, 2007,

2008; Hagiu, 2004b). Public and private services can coexist, with endogenous private fees and participation (Hua et al., 2016; Peitz and Sato, 2026; Liu et al., 2026). Research complementarities and product-market rivalry affect innovation returns, while cross-jurisdictional benefits can separate local provision incentives from national welfare (d'Aspremont and Jacquemin, 1988; Cremer et al., 1997). None of these mechanisms, taken separately, is a novelty claim of this paper.

We ask how private fees and a government's decision to add free basic access depend on regional market incidence, venture-profit appropriation, service demand and technical complementarity, and when decentralised provision agrees with national welfare evaluation. The answer has two deliberately different scopes. Conditional analytical comparisons identify public-cost boundaries when the partner region fully participates. Separate certified examples evaluate all four institutional entry incentives and the national welfare ranking of sixteen provision configurations. The existing model does not deliver a general equilibrium classification for every regional demand, cost or market-structure parameter.

The model has two jurisdictions, each with a welfare-maximising government, a profit-maximising private intermediary and heterogeneous prospective founders. Four independent entry decisions precede private fees and participation. Cross-region pairs undertake research; successful ventures compete with incumbents in differentiated Cournot markets. A founder's technical ability equals her valuation of the premium service, and success includes the product of the partners' abilities. The private premium supplies direct service value rather than a productivity improvement. Governments value regional service benefits, consumers, incumbents and assigned venture earnings; locally paid fees are transfers. The public instrument is service availability, with no choice of buildings, subsidies or procurement.

First, private pricing selects founder composition. Conditional on partner participation, stronger skill complementarity raises the interior private-only fee and cutoff, increasing average participant ability while reducing participation and the total represented skill moment. Free basic access restores coverage alongside a paid premium. It reduces the specialist's conditional fee receipts, but entry remains profitable under the stated sufficient fixed-cost condition. These analytical results organise familiar screening and mixed-provision mechanisms rather than establish a new screening theory.

Second, the conditional public-cost ceiling combines direct service gains, baseline matched-project returns and skill-related commercialisation incidence. Proposition 1 identifies a primitive

profit-share band in which technical complementarity raises that ceiling even though successful ventures have a negative net regional market effect. The local founder share must be large enough to generate private exclusion, yet small enough that consumer and incumbent effects dominate the assigned venture profits. Basic-service value relative to premium value lowers the critical share under the proposition's restrictions. The sign result holds analytically for sufficiently weak positive matching, uniformly on a fixed feasible complementarity interval. A separate exact-rational certificate verifies negative commercialisation incidence, positive access value and a positive value slope on a continuous two-parameter region at the original finite matching intensity. The share band can be written as

$$\omega_i^{\text{crit}} < \omega_i < \min\{1, -M_i/\Pi\}.$$

This is the strongest candidate applied-theory contribution; its independence from all potentially encompassing predecessor theorems is not established by institutional differences alone.

Third, local and national public-cost ceilings differ by the other region's welfare increment. The familiar externality wedge here operates through endogenous fees, selected abilities and regionally assigned commercialisation returns. In the three certified baseline environments, the national welfare maximum over sixteen configurations is always (B, B), where B means basic public access plus private premium provision. Decentralised provision is (B, P) with skill-based success at zero complementarity, (B, B) at complementarity 0.5, and (B, P) with the equal-full-participation-mean pooled technology at 0.5; P means private only. These are exactly certified examples, not a general equilibrium comparative-static theorem. The three-way pattern requires a narrow public-cost band and disappears in the recorded Beta$(2, 2)$ ability alternative. Public overprovision is not established by these examples.

The contribution is therefore a set of economically interpretable, conditional policy boundaries and their bounded equilibrium illustrations. The particular interaction between privately appropriable technological gains, fee-excluded basic services and adverse regional commercialisation is more informative than a count of institutions or results. It does not make private screening, public/private coexistence or interregional externalities new. Related literature and the remaining theorem-absorption uncertainty are discussed in Section 8.

Section 2 presents the primitives. Section 3 derives private pricing and selection. Section 4 characterises regional public-cost ceilings and the central conditional proposition. Section 5

compares local and national evaluations and reports the three four-party provision examples. Section 6 examines equal-mean technologies, continuous certification, economic scale and negative portability results. Sections 7 and 8 discuss measurement, interpretation and prior research, before Section 9 concludes. The Online Supplement retains the derivations and certification boundaries.

2 The model

2.1 Institutions, firms, and timing

There are two jurisdictions $i \in \{A, B\}$. Each contains a local government G_i, a private innovation intermediary P_i, and a population $N_i > 0$ of prospective venture founders. These four institutional decision makers are distinct from incumbent product-market producers. The government maximises its own jurisdiction's welfare; the private intermediary maximises fee revenue net of its real entry cost.

In stage 1 each G_i and P_i simultaneously chooses whether to provide its service. Write $s_i \in \{0, G, P, B\}$ for neither, government only, private only, or both, respectively. There are sixteen profiles $s = (s_A, s_B)$. The government supplies basic innovation intermediation at zero user price, financed by a real fixed cost F_{G_i}. An active private provider offers a premium service and pays real fixed cost F_{P_i}. Public and private services may coexist within a jurisdiction.

In stage 2, active private providers simultaneously post fees $f_i \geq 0$. In stage 3 founders observe available services and choose participation and, where relevant, the premium offering, or an outside option normalized to zero. In stage 4 randomly formed cross-jurisdiction founder pairs invest in a joint research venture. In stage 5 a successful venture competes with operating incumbent producers in each jurisdiction's differentiated-product Cournot market. Founder skills and product demand parameters are common knowledge as distributions, but each founder privately observes her own type before selecting a service.

The stage-1 game concerns availability of *intermediation services*; G_i is not an assertion of publicly owned buildings. Zero-hub incumbent output remains positive. A pair's successful venture serves its own separate product niche, preventing multiple matched projects from repeatedly appropriating the profits of one new product. The maintained tractable specification is deliberately not an unrestricted many-venture Cournot game.

2.2 Successful venture's downstream product market

In market i, let $n_i \geq 1$ identical incumbent producers supply the old product, with aggregate quantity $Q_i = n_i x_i$, and let one successful venture supply a differentiated new good in quantity y_i. The representative consumer's gross quasi-linear surplus is

$$U_i(Q_i, y_i) = a_i Q_i + a_i^v y_i - \frac{b_i Q_i^2 + 2\chi_i Q_i y_i + d_i y_i^2}{2}, \quad b_i, d_i > 0, \quad b_i d_i > \chi_i^2. \tag{1}$$

Taking linear marginal costs into the net intercepts A_i for an incumbent and D_i for the venture, simultaneous Cournot first-order conditions are

$$A_i - b_i(n_i + 1)x_i - \chi_i y_i = 0, \qquad D_i - d_i(2y_i) - \chi_i n_i x_i = 0. \tag{2}$$

Consequently, on the maintained strictly positive interior branch,

$$\mathcal{D}_i = 2b_i d_i(n_i + 1) - n_i \chi_i^2, \quad x_i = \frac{2d_i A_i - \chi_i D_i}{\mathcal{D}_i}, \quad y_i = \frac{b_i(n_i + 1)D_i - n_i \chi_i A_i}{\mathcal{D}_i}. \tag{3}$$

The entrant's combined successful-product operating profit in both markets is $\Pi = \sum_i d_i y_i^2 > 0$. With no venture, incumbents produce $x_i^0 = A_i/[b_i(n_i + 1)] > 0$. The consumer-surplus function evaluated at Cournot quantities is $(b_i Q_i^2 + 2\chi_i Q_i y_i + d_i y_i^2)/2$ and incumbents' operating profits total $n_i b_i x_i^2$.

Define the downstream external effect on consumers and incumbents in region i as

$$\begin{aligned} M_i = & \left[\frac{b_i(n_i x_i)^2 + 2\chi_i n_i x_i y_i + d_i y_i^2}{2} + n_i b_i x_i^2\right] \\ & - \left[\frac{b_i(n_i x_i^0)^2}{2} + n_i b_i (x_i^0)^2\right]. \end{aligned} \tag{4}$$

M_i *excludes* the entrant's Π and is not assumed positive: consumer gains may coexist with incumbent business stealing. $\omega_A, \omega_B \geq 0$, $\omega_A + \omega_B = 1$, divide the venture's private return between the two founders. Market parameters and costs are restricted so that (3), baseline quantities, and the probabilities below are economically feasible; no claims are made outside that domain.

2.3 Complementary skills and research investment

Each region's prospective founders have independently distributed $\theta_i \sim U[0,1]$. In region A, θ_A can represent research and design capability; in region B, θ_B may represent manufacturing or commercial implementation. This is a type interpretation, not a claim that whole geographic economies specialise in those sectors. The perfect alignment of premium-service valuation z_i and technical skill θ_i is analytically convenient but an identifying limitation, not a description of a measured population; public-service users may have lower average technical ability than private-service users without being unproductive. The two regions need not correspond to the spatial boundary of a physical facility.

For a matched pair using research effort $e \geq 0$, project success is

$$p(e, \theta_A, \theta_B) = p_0 + \zeta e + \lambda \theta_A \theta_B, \qquad c(e) = ke^2/2, \quad k > 0, \ \lambda \geq 0. \tag{5}$$

The entire relevant domain satisfies $0 \leq p \leq 1$. Joint investment maximises $p(e, \theta_A, \theta_B)\Pi - ke^2/2$ and yields

$$e^* = \frac{\zeta \Pi}{k}, \quad p_b = p_0 + \frac{\zeta^2 \Pi}{k}, \quad R_0 = p_0 \Pi + \frac{\zeta^2 \Pi^2}{2k}. \tag{6}$$

Optimising effort fixes a common success baseline p_b and net-return intercept R_0, separating gross successful-project effects from real research costs. It creates no independent effort-selection mechanism: the provision contrast works through participant selection and the skill-dependent success term. The net expected venture return, including research cost, equals $R_0 + \lambda \Pi \theta_A \theta_B$. As (5) is separable, the optimal *effort* does not vary with types; the skill channel operates through *success probabilities*, not through an unmodelled type-dependent effort choice.

2.4 Founder selection and matching

The direct basic service value for a founder in region i is

$$u_i(s_j) = v_i + \rho w_i \mathbf{1}\{s_j \neq 0\} > 0, \qquad j \neq i. \tag{7}$$

A private specialist additionally provides $\Delta_i \theta_i$ with $\Delta_i > 0$, and charges fee f_i. This is a direct service-use value, not an improvement in research productivity. Conditional on participating, basic-only users and premium purchasers use the same matching and success technology; premium purchase changes neither p, research cost nor product demand. Specialist technical

advice that directly improves a project is outside this specification. The baseline assumes the premium-service valuation type is perfectly positively associated with technical ability: $z_i = \theta_i$. This maintained assumption is not an empirical claim or a general theory of arbitrary skill–taste correlation.

Define a participation mass and its *unnormalised* first skill moment,

$$m_i = \mathbb{E}[\mathbf{1}\{\text{participates}\}], \qquad q_i = \mathbb{E}[\theta_i \mathbf{1}\{\text{participates}\}]. \tag{8}$$

With pair-matching intensity $\rho\mu > 0$, write $C = \rho\mu N_A N_B$ and

$$a_i^m = \rho\mu N_j \omega_i R_0, \qquad b_i^m = \rho\mu N_j \omega_i \lambda \Pi. \tag{9}$$

These coefficients are *not* the Cournot demand slopes. A private-only founder receives

$$u_i + a_i^m m_j + (\Delta_i + b_i^m q_j)\theta_i - f_i \tag{10}$$

and participates precisely when this expression is nonnegative. A free basic public service is available to all founders; under co-provision the premium upgrade is chosen iff $\Delta_i \theta_i \geq f_i$.

With no local provider, $(m_i, q_i) = (0, 0)$. With G or B, $(m_i, q_i) = (1, 1/2)$. With P only, define

$$\ell_i = \left[\frac{f_i - u_i - a_i^m m_j}{\Delta_i + b_i^m q_j}\right]_0^1, \qquad m_i = 1 - \ell_i, \qquad q_i = \frac{1 - \ell_i^2}{2}, \tag{11}$$

where $[\cdot]_0^1$ truncates to $[0, 1]$. Expected aggregate match volume is $L_0 = C m_A m_B$, while the success-weighted complementarity exposure is $L_1 = C q_A q_B$. They are *distinct endogenous aggregates*; even equal $m_A m_B$ need not imply equal $q_A q_B$.

2.5 Regional welfare and feasible public choice

Let W_i^0 be consumer plus incumbent producer surplus in the no-venture Cournot market. For a successful venture, define

$$K_i = M_i + \omega_i \Pi, \quad S_i^0 = p_b M_i + \omega_i R_0, \quad S_i^1 = \lambda K_i. \tag{12}$$

The gross real service benefit is $Z_i = 0$ with no provision; $Z_i = N_i u_i$ with G; $Z_i = N_i(u_i m_i + \Delta_i q_i)$ with P; and $Z_i = N_i(u_i + 3\Delta_i/8)$ with B (using the $f_i = \Delta_i/2$ premium continuation below).

Local public objectives and active private provider profits are

$$W_i(s) = W_i^0 + Z_i(s) - F_{G_i}\mathbf{1}\{G_i \in s_i\} - F_{P_i}\mathbf{1}\{P_i \in s_i\} + Cm_A(s)m_B(s)S_i^0 + Cq_A(s)q_B(s)S_i^1, \tag{13}$$

$$\pi_i^P(s) = N_i f_i(s) D_i(s) - F_{P_i}\mathbf{1}\{P_i \in s_i\}, \quad D_i = \begin{cases} m_i, & s_i = P, \\ 1/2, & s_i = B. \end{cases} \tag{14}$$

Absent private providers have payoff zero. The real cost of a local private service provider is included in local welfare. Founders' fees are intraregional transfers to local private intermediaries and must *not* be subtracted again from (13).

Each government chooses only whether to supply its free basic intermediation. For comparison we report the *restricted-instrument* two-jurisdiction benchmark

$$s^{RP} \in \underset{s \in \{0,G,P,B\}^2}{\arg\max} \{W_A(s) + W_B(s)\}, \tag{15}$$

holding downstream fee, participation, research and Cournot continuations decentralised. This is not a first best, nor an optimisation over subsidies, digital-only systems or construction contracts.

3 Private intermediation and founder selection

Private fees determine which founders obtain basic access and which purchase specialist services. We first separate conditional monopoly pricing from the coupled private/private fee game. These familiar selection mechanisms provide the continuation allocations used in the public-provision and welfare comparisons below.

3.1 Private fee and participation choices

Fix participation in the other region and define $U_i = u_i + a_i^m m_j > 0$, $T_i = \Delta_i + b_i^m q_j > 0$. For $s_i = P$, fees and selection satisfy

$$\ell_i = [(f_i - U_i)/T_i]_0^1, \qquad \mathcal{R}_i(f_i) = N_i f_i (1 - \ell_i). \tag{16}$$

The global fee optimum on the nonnegative half-line, including the full-participation kink, is

$$(f_i^*, \ell_i^*) = \begin{cases} ((U_i + T_i)/2,\ (T_i - U_i)/(2T_i)), & 0 < U_i < T_i, \\ (U_i,\ 0), & U_i \geq T_i. \end{cases} \tag{17}$$

In particular, one must not extend the interior optimum through the kink. If $s_i = B$, free participation is universal and the premium upgrade demand is $N_i(1 - f_i/\Delta_i)$ for $f_i \in [0, \Delta_i]$. Hence

$$f_i^B = \Delta_i/2, \quad D_i^B = 1/2, \quad \mathcal{R}_i^B = N_i\Delta_i/4. \tag{18}$$

When $(s_A, s_B) = (P, P)$, neither region's participation moments may be held constant while optimising its private fee. A candidate price pair must jointly solve

$$\ell_i = \left[\frac{f_i - u_i - a_i^m(1 - \ell_j)}{\Delta_i + b_i^m(1 - \ell_j^2)/2} \right]_0^1, \quad i = A, B, \tag{19}$$

alongside each provider's global nonnegative-fee best response to f_j, allowing both boundary and interior fee branches. On the interior branch the stationary equations may be written in terms of the coupled participation system; stationarity alone is *not* a globally valid fee-equilibrium certificate. The Online Supplement proves a baseline-specific global pure-price equilibrium result using analytical best-response bounds and an exact rational interval certificate, for the three reported numerical environments only. Off-path cases outside those environments remain subject to separate validation.

A useful distinction: coexistence is not a private-revenue complementarity. In the conditional comparison between a private-only service and a private premium offered alongside free basic access, hold the other jurisdiction's participation moments fixed. With $U_i > 0$ and $T_i = \Delta_i + b_i^m q_j \geq \Delta_i$, the gross receipts of the local private provider are

$$\mathcal{R}_i^P = \begin{cases} N_i(U_i + T_i)^2/(4T_i), & U_i < T_i, \\ N_i U_i, & U_i \geq T_i, \end{cases} \qquad \mathcal{R}_i^B = N_i\Delta_i/4. \tag{20}$$

Both branches imply $\mathcal{R}_i^P > \mathcal{R}_i^B$. In the first branch, $(U_i + T_i)^2/T_i > T_i \geq \Delta_i$; in the second, $U_i \geq T_i \geq \Delta_i$. Thus adding free public basic service *strictly displaces private fee receipts* under these conditional primitives. In fact, under the maintained positive basic benefit and

nonnegative matching returns, $F_{P_i} < N_i\Delta_i/4$ is sufficient for the private intermediary's entry to be a strictly dominant first-stage action: co-provision yields this gross premium revenue, while private-only provision can secure strictly more. Both benchmark private fixed costs satisfy this condition. The coexistence studied here is therefore a mixed provision configuration, *not* a theorem that the government's entry raises the private provider's earnings. With the original region-B parameters, gross premium revenue in B is 0.25 and net private profit is $0.01 > 0$. This transparent distinction prevents a policy interpretation that confuses co-location/coexistence with profit complementarity.

3.2 Technical complementarity and conditional selection

The original lemma identifiers are retained for the Online Supplement: the pricing result, Lemma 2, precedes the incidence result, Lemma 1, in the economic sequence.

Lemma 2 (conditional fee selection). Fix the other region's (m_j, q_j) with $q_j > 0$, maintain $0 < U_i < T_i$, $\omega_i > 0$, $\Pi > 0$ and $z_i = \theta_i$. Increasing λ then implies

$$\frac{\partial f_i^*}{\partial \lambda} > 0, \quad \frac{\partial \ell_i^*}{\partial \lambda} > 0, \quad \frac{\partial m_i}{\partial \lambda} < 0, \quad \frac{\partial q_i}{\partial \lambda} < 0, \quad \frac{\partial (q_i/m_i)}{\partial \lambda} > 0. \tag{21}$$

Indeed $T_i' = \rho\mu N_j \omega_i \Pi q_j > 0$, while U_i is constant, and the monopoly formula gives $\ell_i^* = 1/2 - U_i/(2T_i)$ and $\ell_i' = U_i T_i'/(2T_i^2) > 0$. Participant mean ability $(1 + \ell_i)/2$ rises even as the total represented ability falls. This is not a P/P price-equilibrium comparative static.

4 Public provision and regional economic characteristics

The policy question is whether the real fixed cost of adding free basic access is justified by the change in regional welfare. The relevant regional characteristics are the consumer and incumbent effect, the share of venture profits retained by founders, and basic-service value relative to premium value. They enter a conditional comparison with the partner region fully participating; no general entry comparative static is assumed.

4.1 Regional commercialisation incidence

Lemma 1 (regional incidence). With full founder participation, fixed interior Cournot quantities and independent uniform abilities,

$$\frac{\partial W_i}{\partial \lambda} = \frac{C}{4} K_i, \qquad \frac{\partial (W_A + W_B)}{\partial \lambda} = \frac{C}{4}(M_A + M_B + \Pi). \tag{22}$$

Local and aggregate signs can differ because incumbent displacement and assigned venture earnings differ across regions. The derivatives follow from $q_A q_B = 1/4$ and $K_A + K_B = M_A + M_B + \Pi$. No participant or price response is included in this full-coverage calculation.

4.2 The conditional public cost ceiling

Lemma 3 (conditional public addition). Fix the other region at $(m_j, q_j) = (1, 1/2)$ and let region i have a private-only continuation with $\ell_i \in (0, 1)$. Define the direct-service gain before government cost by

$$H_i = N_i \left(u_i \ell_i - \frac{\Delta_i}{8} + \frac{\Delta_i \ell_i^2}{2} \right). \tag{23}$$

Adding free basic access while retaining the private specialist is locally preferred exactly when $F_{G_i} \leq F_i^{\text{local}}$, where

$$F_i^{\text{local}} = H_i + C \left(S_i^0 \ell_i + \frac{\lambda K_i \ell_i^2}{4} \right). \tag{24}$$

The national comparison uses

$$F_i^{\text{national}} - F_i^{\text{local}} = C \ell_i \left(S_j^0 + \frac{\lambda K_j \ell_i}{4} \right). \tag{25}$$

This foreign increment may be positive, zero or negative.

To verify the result, public access raises local mass by ℓ_i and local ability moment by $\ell_i^2/2$; with the opposite moment $1/2$, skill exposure rises by $C\ell_i^2/4$. The direct-service change is H_i and private real costs cancel. Adding the neighbouring welfare increment gives the national threshold. The same accounting extends to full-coverage corners using the realised cutoff. It does not establish a full four-entry equilibrium or an optimum over fees, research allocations, subsidies or buildings. The Online Supplement provides the continuation and accounting proofs.

4.3 Regional primitives and technical complementarity

Consider the public-access margin from private-only provision (P) to free basic access coexisting with private premium provision (B). Fix the partner jurisdiction in G or B, so $(m_j, q_j) = (1, 1/2)$; the private intermediary in region i nevertheless reoptimises its fee in the private-only continuation. This is a *conditional institutional comparison*, not a global comparative static of all four stage-1 entry decisions.

Write $d = \Delta_i$, $\gamma = \rho\mu$, $C = \gamma N_i N_j$, and set

$$U = u_i + \gamma N_j \omega_i R_0, \quad \eta_i = \frac{\gamma N_j \omega_i \Pi}{2}, \quad D = d + \eta_i \lambda, \quad \ell = \frac{D - U}{2D}, \tag{26}$$

on the interior branch $0 < U < D$. The private-only fee is $(U + D)/2$, whereas free basic provision restores access to all founder types, with premium pricing at $d/2$. Using Lemma 3, the gross local public-access value, before government fixed cost, is

$$T_i(\lambda) = N_i \left(u_i \ell - \frac{d}{8} + \frac{d\ell^2}{2} \right) + C \left(S_i^0 \ell + \frac{\lambda K_i \ell^2}{4} \right), \qquad K_i = M_i + \omega_i \Pi. \tag{27}$$

Differentiating while keeping the research-market primitives, ρ, and u_i fixed gives the exact identity

$$\begin{aligned} T_i'(\lambda) = {} & \left[N_i(u_i + d\ell) + C S_i^0 \right] \frac{U \eta_i}{2D^2} \\ & + \frac{C K_i}{4} \left(\ell^2 + \frac{\lambda \ell U \eta_i}{D^2} \right). \end{aligned} \tag{28}$$

The first line describes the access restored by the rising private fee-induced exclusion cutoff, including its baseline matching component; the second is the regional return to the technically different additional research pairs. The two need not have the same sign.

Proposition 1 (the screening–incidence inversion). Maintain the model's uniform and perfectly aligned technical-skill/premium-value distribution, a positive interior Cournot continuation, an interior local private-only fee, partner full participation, $M_i < 0$, $0 < u_i < d$, $\Pi > 0$, and nonnegative R_0. Fix a compact interval of technologically feasible $\lambda \geq 0$. Define

$$\omega_i^{\text{crit}} = \frac{-M_i}{\Pi} \frac{(d - u_i)^2}{d^2 + 3u_i^2}. \tag{29}$$

Suppose the interval

$$\boxed{\omega_i^{\text{crit}} < \omega_i < \min\left\{1, \frac{-M_i}{\Pi}\right\}} \tag{30}$$

is nonempty. As $\gamma = \rho\mu \downarrow 0$, holding ρ and the basic-service value u_i fixed, the conditional public-access threshold satisfies, uniformly on the specified λ interval,

$$\frac{\partial T_i}{\partial \lambda} = \frac{\gamma N_i N_j}{16 d^2}\left[M_i(d-u_i)^2 + \omega_i \Pi(d^2 + 3u_i^2)\right] + O(\gamma^2). \tag{31}$$

Consequently, at every sufficiently small positive γ,

$$K_i < 0, \qquad S_i^0 < 0, \qquad \frac{\partial T_i}{\partial \lambda} > 0 \tag{32}$$

throughout the λ interval. If in addition $0 < u_i < 2d/3$, then $T_i(\lambda) > 0$ uniformly at sufficiently small positive γ. Thus public provision can have a positive affordable fixed cost and become more valuable as technical complementarity rises even though its incremental research-commercialisation surplus is strictly negative in the financing region.

The critical share ω_i^{crit} strictly decreases in u_i/d on $(0,1)$. At $\omega_i = 0$ and $M_i < 0$, by contrast, the private screening response to λ disappears and $T_i'(\lambda) = CK_i\ell^2/4 < 0$ on the interior branch. These are implications of the existing primitives rather than new policy instruments. The proof and the uniformity conditions are in the Online Supplement. The proposition concerns the *gross conditional public cost ceiling*, not an all-parameter four-player subgame-perfect equilibrium or an optimum across taxes, subsidies and physical facilities.

Two distinctions matter. First, $S_i^0 = p_b K_i - \omega_i \zeta^2 \Pi^2/(2k) < 0$ when $K_i < 0$; the adverse regional return applies to baseline as well as skill-complementarity-induced ventures. Second, (30) provides a nonempty feasible band only when $\omega_i^{\text{crit}} < \min\{1, -M_i/\Pi\}$; positive basic-service value alone does not guarantee that its intersection with feasible profit shares is nonempty. The sign restriction is first-order in weak matching, while Section 6 separately certifies a finite-matching closed rectangle for the original economic environment.

What regional characteristics identify. The market effect M_i aggregates consumer and incumbent changes derived from the existing Cournot demand, cost and incumbent-count primitives; it is not an independent urban or rural indicator. The share ω_i assigns joint-venture profits to the region's founder. Proposition 1 identifies an intermediate share: large enough for private technological returns to increase exclusion, but small enough that incumbent and

consumer losses still dominate locally. The ratio u_i/Δ_i describes the value of free basic access relative to specialist services. Its proved effect is a decrease in the critical profit share on the stated interior domain, not a general monotonicity theorem for the level of public value.

Candidate populations N_i, N_j and matching intensity $\rho\mu$ enter both private willingness to pay and the scale of regional matching surplus. The leading weak-matching slope displays their scale explicitly, holding the remaining primitives fixed. It does not prove that larger populations or stronger matching always raise the public cost ceiling or change the four-party equilibrium. Similarly, the Cournot formulas identify how market structure feeds into M_i and Π, but the paper has no general signed comparative static for every demand, cost or incumbent-count change. The scope of the research question is therefore conditional analytical boundaries plus certified provision examples, rather than a complete classification of heterogeneous regional economies.

Reading the strict baseline-incidence claim. The Online Supplement's proof of the strict inequality for S_i^0 explicitly uses a positive baseline success intercept $p_b > 0$. The numerical baseline satisfies that condition. The statement above is retained from the frozen version; its strict baseline-incidence conclusion should be read with that proof condition visible. If both p_0 and ζ are zero, the baseline term is zero rather than strictly negative. This inherited scope issue does not change the fee-selection slope or the certified finite-matching example, but it prevents treating the printed statement as a theorem on the degenerate zero-baseline case.

5 Decentralised provision and social welfare

Lemma 3 supplies separate local and national cost ceilings. The difference in equation (25) is the welfare increment accruing to the other region through additional matches and their skill composition. When that increment is positive, a public cost between the two ceilings makes local non-provision consistent with a national preference for adding access; a negative increment reverses the conditional ranking. Equality of the ceilings removes this particular wedge. These are comparisons of the same two configurations, not a proof that the full decentralised and restricted-planner choices coincide whenever that increment vanishes.

The externality accounting is familiar. Here its economic content comes from the fee-selected cutoff, technical success and the allocation of downstream venture gains between founders, consumers and incumbent producers. A local threshold is only one incentive in the four-party entry

game. We therefore apply the separate entry and continuation checks before drawing conclusions about a provision configuration. No four-party example of public overprovision is certified.

5.1 First-stage equilibrium and planner evaluation

For each of the sixteen entry configurations, let $\mathcal{E}(s)$ denote the set of admissible downstream equilibria. Where a continuation is selected, assign the four stage-1 payoffs from (13) and (14). A candidate pure first-stage Nash equilibrium (NE) must make every binary entry flip weakly unprofitable:

$$W_i(s) \geq W_i(s_i^{G\text{-flip}}, s_j) \qquad \text{for each } G_i, \tag{33}$$

$$\pi_i^P(s) \geq \pi_i^P(s_i^{P\text{-flip}}, s_j) \qquad \text{for each } P_i. \tag{34}$$

The four states in one region are $0 \leftrightarrow G$ and $P \leftrightarrow B$ for a government flip, and $0 \leftrightarrow P$ and $G \leftrightarrow B$ for a private flip. Thus there are four independent binary incentive inequalities at each state; a government's entry comparison does *not* subsume the private provider's response. Equilibrium ties may generate multiplicity. The restricted planner in (15) compares all sixteen total-welfare values *under the same allowed continuation rule*. Without an admissible off-path equilibrium one cannot declare a global first-stage SPNE.

The selected baseline cases use an analytic price-best-response argument and exact rational enclosure of the coupled (P, P) equilibrium. The Online Supplement documents the global deviation bounds, the continuation fee boxes, and the strict first-stage incentive signs for each baseline. This is neither a general uniqueness theorem for all primitives nor a complete Lean formalisation of the entire sequential game.

5.2 Three certified provision configurations

We retain the original numerical primitives rather than selecting a wider reversal after the fact. All magnitudes are model units, not an empirical calibration of a particular programme. The baseline has $(n_A, n_B) = (5, 2)$, $(N_A, N_B) = (3, 1)$, $\Delta_A = \Delta_B = 1$, $\rho = 0.4$, $\mu = 0.25$, $(\omega_A, \omega_B) = (0.8, 0.2)$, $(F_{G_A}, F_{G_B}) = (0.09, 0.0405)$, $(F_{P_A}, F_{P_B}) = (0.65, 0.24)$, $p_0 = 0.05$, $\zeta = 0.005$, $k = 0.0004$, $\chi_i = 0.9$, $(v_A, v_B) = (0.15, 0.10)$ and $(w_A, w_B) = (0.22, 0.45)$; the Cournot net intercepts and own-good slopes equal one. The pooled comparison replaces $\lambda\theta_A\theta_B$ by $\lambda/4$ throughout individual returns and welfare, preserving the mean under complete participation

only.

Table 1 reports the three original outcomes. For each displayed setting, the P/P continuation is a unique global pure-price equilibrium by an analytic best-response argument with rational interval verification. All sixteen entry profiles and four unilateral flips per profile are checked. These certificates cover the three settings and the explicitly stated small perturbation region, not arbitrary parameters or a continuum of complementarity values.

[Table 1 about here.]

6 Technology heterogeneity, equilibrium scale and robustness

The preceding provision examples connect local and national decisions. We now separate their technological mechanism, the continuous conditional sign certificate, and the much narrower domain on which the full provision configurations persist.

6.1 Fixed participation and selected abilities

Corollary 1 (the fixed-mass benchmark). At participation mass $m_i = 1 - \ell_i$, selected ability exceeds random admission of the same mass by

$$q_i - \frac{m_i}{2} = \frac{\ell_i(1-\ell_i)}{2} \geq 0. \tag{35}$$

At a fixed cutoff and with the other region fully covered, the selected-minus-pooled technological welfare increment is

$$(W_i^{\text{skill}} - W_i^{\text{pooled}})\Big|_{\text{same cutoff}} = \frac{C\lambda K_i}{4}\ell_i(1-\ell_i). \tag{36}$$

It has the sign of K_i; it does not compare reoptimised pricing equilibria.

Exact participation reduction. Eliminating the uniform upper-tail cutoff yields

$$q_i = \phi(m_i), \quad \phi(m) = m - m^2/2, \quad q_A q_B = \phi(m_A)\phi(m_B). \tag{37}$$

The identity also holds in $0, G, B$ states. Match volume and ability-weighted exposure have different economic meanings, but are not independent endogenous state dimensions here. A nonlinear participation-externality representation is possible.

6.2 Equal-mean homogeneous success and repriced founder selection

A related diagnostic explains why matching technologies with equal full-population mean success rates can imply different values of public access. Compare the heterogeneous premium $\lambda\theta_i\theta_j$ (H) with a pooled homogeneous premium $\lambda/4$ (L). With the same partner fully participating, write $h = \eta_i\lambda$ and retain U from (26). Their respective globally optimal private-only cutoffs, including the full-coverage corners, are

$$\ell_H = \max\left\{0, \frac{d+h-U}{2(d+h)}\right\}, \qquad \ell_L = \max\left\{0, \frac{d-U-h/2}{2d}\right\}. \tag{38}$$

The pooled technology increases the private-only participation intercept by $h/2$ and removes the slope h; the two comparisons must not be evaluated at the same unoptimised fee. For $h > 0$, $\ell_H \geq \ell_L$, with strict inequality unless both cutoffs are zero. When both are interior,

$$\ell_H - \ell_L = \frac{h(d+h+2U)}{4d(d+h)} > 0. \tag{39}$$

Let T_H and T_L denote the public cost ceilings for the heterogeneous and equal-mean pooled technologies after optimal fee readjustment. With $\delta = \ell_H - \ell_L$, substitution into Lemma 3 gives

$$\begin{aligned} T_H - T_L = \delta\left[N_i\left(u_i + \frac{d}{2}(\ell_H + \ell_L)\right) + CS_i^0\right] \\ + \frac{C\lambda K_i}{4}(\ell_H^2 - \ell_L). \end{aligned} \tag{40}$$

This holds also at the full-participation corners. The first line measures restored basic-service value and baseline matched-project incidence when privately selected founder exclusion changes. The second compares different skill-weighted exposure of restored projects. Its sign can differ from the first line, so even equal full-participation mean project success need not imply equal equilibrium access value. If a fixed government cost lies strictly between T_H and T_L, the conditional public-entry decision reverses; this alone does not certify the global four-entrant equilibrium.

The comparison can also reverse in the other direction. With $\omega_i = 0$, $M_i > 0$ and $0 < u_i < d$, the two technologies induce the same positive cutoff ℓ, but $T_H - T_L = C\lambda M_i\ell(\ell-1)/4 < 0$ for $\lambda > 0$. The strict sign persists for sufficiently small positive ω_i by continuity. A positive technical effect on local product-market welfare can therefore diminish the marginal value of restoring a lower-skill group. The proposition and this equal-mean diagnostic answer different questions:

Proposition 1 changes λ within the heterogeneous technology; (40) changes the distribution of technical gains at fixed λ. Neither yields an unrestricted policy prescription.

6.3 The conditional proposition at finite matching intensity

The principal analytical result is not the discrete three-scenario equilibrium switch; it is the sign of the conditional public-access cost ceiling when skill complementarity increases. At the original $p_0 = 0.05$, $\gamma = 0.1$, $u_B = 0.28$ and the original differentiated-product Cournot solution, the weak-matching primitive-share boundary from Proposition 1 is

$$\omega_B^{\text{crit}} \simeq 0.1167164, \qquad -M_B/\Pi \simeq 0.2781020.$$

The original $\omega_B = 0.2$ is strictly between them. Unlike this weak-matching asymptotic criterion, the following finite-γ validation is independent of taking γ arbitrarily close to zero.

The Online Supplement's exact rational interval checker proves that, for *every* (ω_B, λ) in

$$0.13 \leq \omega_B \leq 0.27, \quad 0 \leq \lambda \leq 0.5, \quad \omega_A = 1 - \omega_B, \tag{41}$$

holding all other original primitive values and full founder participation in A, the B-region continuation is interior and satisfies

$$K_B < 0, \quad S_B^0 < 0, \quad H_B > 0, \quad T_B(\lambda) > 0, \quad \frac{\partial T_B}{\partial \lambda} > 0, \quad C\left(S_B^0 \ell_B + \frac{\lambda K_B \ell_B^2}{4}\right) < 0. \tag{42}$$

All strict inequalities are proved over the *continuous* two-dimensional rectangle using inclusion-bounded exact rational subrectangles, not from a grid of simulated equilibrium outcomes. The certified conservative lower bounds are $T_B > 0.04036779$ and $\partial T_B/\partial \lambda > 0.00003340$; the venture-related increment is strictly negative throughout. This establishes a genuinely nondegenerate primitive domain for the *conditional* screening–incidence inversion at finite matching intensity. It does *not* imply that the full four-entrant pure equilibrium changes at all these parameter values.

With original $\omega_B = 0.2$, the conditional cost ceiling rises from $T_B(0) \simeq 0.0404250710$ to $T_B(0.5) \simeq 0.0405316701$, and the derivative is strictly positive for every intermediate λ. Despite its continuously verified sign, the entire increase is only 0.0001065991 in model cost units. A fixed cost must lie between the two ceilings for the conditional public-entry decision to change;

proving the actual four-entrant stage-1 switch additionally requires the separate off-path Nash and entry-deviation certificates reported in Section 5.

What drives the three-way switch? In region B, $M_B \simeq -0.0307750$ and $\omega_B \Pi \simeq 0.0221322$, so $K_B \simeq -0.00864284 < 0$ and $S_B^0 \simeq -0.000568455 < 0$. Additional successful ventures have a negative net local market contribution, including the founders' assigned return. Table 2 decomposes its public-entry threshold into direct service, baseline matching and skill-related success incidence. Free access is locally attractive in the skill-0.5 case because its direct-service value exceeds the two negative venture terms and the public cost. The result is not driven by a positive local innovation externality.

[Table 2 about here.]

Increasing complementarity from zero to 0.5 changes the private cutoff by only 0.0002322968. The resulting 0.0001486557 increase in the direct-service term is offset by a 0.0000420170 skill-success loss and a 0.0000000396 baseline matching loss. Their net is 0.0001065991, exactly the width of the three-scenario public-cost band. The relevant strength in the fee problem is $h = b_B^m/2 \simeq 0.00165991$, small relative to $\Delta_B = 1$; hence complementarity barely shifts private exclusion in this calibration. Small threshold displacement, followed by an indicator decision near indifference, explains both the configuration change and its limited economic magnitude.

At $\lambda = 0.5$, the heterogeneous–pooled threshold gap is larger, 0.0004884409. Equation (40) attributes it to 0.0004139716 from changed access and baseline matching plus 0.0000744693 from technological incidence. Because $K_B < 0$, the lower technical exposure of the additional selected collaborations makes public access less locally costly than in the pooled comparison. This is a regional-incidence effect with repriced participation, not an assertion that lower-ability founders are intrinsically more innovative.

Cost scale and verification domain. Opposite heterogeneous and pooled decisions at $\lambda = 0.5$ require

$$0.040043229233 < F_{G_B} < 0.040531670102. \tag{43}$$

Retaining all three equilibria additionally requires

$$\begin{gathered} \underline{F}_B^{(3)} < F_{G_B} < \overline{F}_B^{(3)}, \\ \underline{F}_B^{(3)} \simeq 0.040425071007120596, \quad \overline{F}_B^{(3)} \simeq 0.04053167010244885. \end{gathered} \tag{44}$$

The endpoints are exact rational indifference values; displayed decimals are approximations. The latter interval is only 0.2632% of the baseline cost, with an upper slack of 0.0782%. An all-sixteen-profile rational certificate establishes the three equilibria and the restricted planner throughout this open cost interval, with only the identified government ties at the outer endpoints. It also verifies a small joint rectangle,

$$\begin{aligned} &0.089999 < F_{G_A} < 0.090001, \\ &0.040499 < F_{G_B} < 0.040501, \\ &0.199999 < \omega_B < 0.200001, \quad \omega_A = 1 - \omega_B, \end{aligned} \tag{45}$$

including its global P/P continuations. These are persistence certificates, not evidence of broad policy relevance.

The conditional government margin has a numerically estimated switch near $\lambda^\star \simeq 0.351184$. Numerical continuation suggests the corresponding first-stage configuration change; the three benchmark P/P certificates do not certify every off-path price game over the intervening λ interval. No continuous-interval full-SPNE or uniqueness claim at the tie is made. The neighbouring region's public-addition gains are positive in the three displayed settings (approximately 0.000414, 0.000768 and 0.001394), explaining why local non-entry can coexist with the restricted planner's preference for (B, B).

6.4 Negative portability and interpretation limits

The existing 396-cell sensitivity panel does not establish broad robustness. At $\Delta_B = 0.8$ or 0.9, or $N_B = 0.8$, its selected numerical continuations yield multiple pure equilibria (B, G) and (B, P) and a planner preference for (B, P). At product substitutability $\chi = 0.95$, the $\lambda = 0.5$ baseline returns to (B, P). These are alternative numerical regimes, not proofs of price equilibria for every perturbed primitive.

Under the separately checked Beta$(2, 2)$ ability distribution, the numerical equilibrium and restricted planner allocation are (P, P) at both $\lambda = 0$ and 0.5. The corresponding conditional region-B public-addition margins are about -0.0305626 and -0.0304756. The original configuration switch therefore disappears at the unchanged costs. The recorded differentiated-Bertrand adverse test is also negative, but its full alternative model has not been independently reconstructed in this revision; no freshly replicated Bertrand result is claimed. Replication files

distinguish these negative tests from the exact baseline certificates.

The maintained equality $z_i = \theta_i$ equates willingness to pay for specialist services with relevant technical ability. Selection on willingness to pay need not identify technological selection in data. The specialist has no direct effect on the success technology, and endogenous effort adds no ability-dependent investment channel. Physical location, building capital, service quality choice, subsidies, taxation, procurement and remote alternatives are outside the policy menu. These restrictions limit interpretation even where the baseline equilibrium is mathematically certified.

7 Empirical implications and policy interpretation

The central policy proposition distinguishes the region's *commercialisation incidence* $K_i = M_i + \omega_i \Pi$ from its willingness to pay for restored basic service, T_i. Empirically, a counterintuitive screening–incidence inversion would require three separate facts: (i) higher skill complementarity is associated with a higher privately optimal fee and greater exclusion among otherwise eligible firms; (ii) venture launch and competition produce a negative combined regional consumer–incumbent–assigned-founder market effect K_i; and (iii) the marginal welfare value of basic service access is sufficiently high that the public-access cost ceiling rises. A descriptive project count cannot verify any of these signs. The primitive inequality (30) additionally requires identifying the local appropriation share ω_i, the net incumbent/consumer effect M_i, and the ratio u_i/Δ_i; these are economic constructs, not quantities directly recorded in a programme dashboard. The weak-matching limit and uniform-ability assumption are maintained theory restrictions, not empirically established for any jurisdiction.

Although the paper is theoretical, its mechanism can be confronted with data on fees, applicants, matched ventures and downstream markets. The distinction between *conditional model predictions* and *causal estimates* is essential. The following three implications follow from existing equations; they are neither additional general equilibrium theorems nor claimed findings from newly collected data. In particular, posted prices, access to basic services and the availability of successful partners are jointly endogenous outside the stated conditional comparisons.

Implication 1: private fees select participant composition as well as participation volume. Hold the other region's moments (m_j, q_j), the service's nonprice characteristics and the local coefficients $U_i > 0$ and $T_i > 0$ fixed. For a private-only service on its interior cutoff

branch, $\ell_i = (f_i - U_i)/T_i$, $m_i = 1 - \ell_i$ and $q_i = (1 - \ell_i^2)/2$. Consequently,

$$\frac{\partial m_i}{\partial f_i} = -\frac{1}{T_i}, \qquad \frac{\partial q_i}{\partial f_i} = -\frac{\ell_i}{T_i}, \qquad \frac{\partial (q_i/m_i)}{\partial f_i} = \frac{1}{2T_i} > 0, \quad 0 < \ell_i < 1. \tag{46}$$

Thus an exogenous fee increase reduces founder participation and the *total* represented skill moment, while raising participants' *average* technical ability. The last effect does not establish an increase in the number of successful projects. A possible empirical test would link the full universe of eligible firms to fee offers, take-up decisions, *pre-service* assessments of technical ability, and subsequent collaborations. After normalising a defensible ex-ante ability measure, one could separately estimate take-up m_i, participant mean skill q_i/m_i, and the skill moment q_i, rather than using registration or match counts as sufficient proxies for innovation quality.

Fee endogeneity is the principal threat: specialists may raise prices because expected demand or client skill improved; targeted discounts may also depend on unobserved ability. A credible design needs plausibly exogenous fee variation (for example, an independently determined tariff change) and a known eligible denominator, not merely a positive correlation between fees and average participant quality. If willingness to pay is poorly aligned with the technical skill relevant to the venture, the model's $z_i = \theta_i$ assumption fails and the positive composition prediction is not warranted. The evidence from European Digital Innovation Hubs that supported client firms can be relatively capable (Carpentier et al., 2026) reinforces the importance of measuring *who participates*; it does not identify this model's fee-selection mechanism.

Implication 2: public basic access may reduce private fee receipts without eliminating a specialist. With an active private provider, a free basic public option and the uniform $z_i = \theta_i$ benchmark, the premium is selected by $\theta_i \geq f_i/\Delta_i$. Its optimal price and take-up share are $f_i^B = \Delta_i/2$ and $D_i^B = 1/2$, and gross fee receipts equal $N_i\Delta_i/4$ (equation (18)). Relative to the private-only fee optimum, with the *other jurisdiction's participation held fixed*, equation (20) implies $\mathcal{R}_i^P > \mathcal{R}_i^B$. The private provider nevertheless remains viable under co-provision when $F_{P_i} < N_i\Delta_i/4$. These are predictions of a particular service differentiation and cost structure, not a universal claim that public programmes displace private firms.

An evaluation should track separately the universe served by free basic services, paid premium upgrades, actual paid fees (net of vouchers), provider receipts and fixed operating costs, and exits or continued operation. A decline in premium revenue does not imply exit. Comparing different jurisdictions before and after public opening without accounting for changes

in specialist quality, local demand, rival services, subsidies and fee policies would not isolate the conditional displacement effect. The prediction of a one-half premium upgrade rate is especially restrictive: it is tied to uniform willingness to pay, linear incremental premium value, one provider and the maintained absence of capacity/quality choice, so it is a useful *benchmark falsification* rather than a likely population constant.

Implication 3: local and cross-region returns to additional intermediation need not align. The project-level complementarity mechanism has a directly falsifiable component: at a given research effort e and within the maintained interior success-probability domain, equation (5) implies

$$\frac{\partial^2 p(e, \theta_A, \theta_B)}{\partial \theta_A \, \partial \theta_B} = \lambda \geq 0. \tag{47}$$

When $\lambda > 0$, an increase in one partner's ex-ante technical capability raises the success-probability increment associated with the other partner's capability. This is a *test of the maintained success technology*, not a novel theoretical theorem. It would require ex-ante skill measures for *both* partners, recorded unsuccessful as well as successful collaborations, effort and project-characteristic controls, and a credible strategy for endogenous pairing and project selection. A positive observed interaction coefficient alone is not proof of causal complementarity.

Hold the neighbouring region j at full founder participation, and compare region i's private-only service with public/private coexistence under their respective equilibrium fee continuations. Lemma 3 implies that the neighbouring jurisdiction's additional surplus—which the local government financing region i does not internalise—is

$$\Delta W_j = C\left(S_j^0 \ell_i + \frac{\lambda K_j}{4} \ell_i^2\right) = F_i^{\text{national}} - F_i^{\text{local}}, \tag{48}$$

where $K_j = M_j + \omega_j \Pi$. The sign of this *foreign* benefit is not fixed: the consumer and founder returns from a new venture may be offset by business stealing from incumbent producers. The empirically discriminating question is therefore whether increases in cross-region matches or partner quality are accompanied by gains or losses *where the new product competes*, not merely where its participating founders are registered.

A feasible data architecture would link eligible firms and their ex-ante skills, pair formation across jurisdictions, project success and product launch, product-market prices and quantities, incumbent performance, founder profit shares, programme budgets and tax-financed real costs.

One must also estimate consumer surplus and the counterfactual outcomes of both markets *without the venture*; aggregate collaboration statistics or participant turnover alone do not identify M_j or either regional welfare threshold. Matched-pair projects and their sequential contracting milestones are observable in some intermediary datasets (Eldebo and Hjelm, 2025), but their existence does not supply the necessary product-market counterfactual. Cross-border demand shocks, endogenous partner choice, market-entry selection and regional trade flows complicate identification.

A disciplined policy use. These implications motivate a diagnostic sequence, not an unestimated cost–benefit claim. First determine *which* firms are reached and which technical skills they bring. Second distinguish basic-service access, private upgrade spending and supplier survival. Third identify where the commercialised products create consumer benefits, founder income and incumbent losses. Only after observing or credibly estimating these components could a government attempt the local-versus-aggregate welfare calculation. The narrow baseline ownership switch reported in Section 6 can discipline a calibrated scenario analysis, but is *not* a sign prediction for an average treatment effect of public hub openings. The model does not compare physical facilities with remote services, vouchers or procurement contracts, and none of these implications establishes that a new public building should be constructed.

8 Related literature and the contribution boundary

Private matchmaking, screening, and heterogeneous network value. Damiano and Li (2007) study an optimising monopoly matchmaker that posts type-contingent entrance-fee schedules to sort privately informed agents into complementary random bilateral matches; Damiano and Li (2008) analyse matching-market competition and the potentially inefficient sorting it produces. These papers already combine *fee-induced selection with complementary match value*, so neither the product of two founder abilities nor price-dependent sorting is a new mechanism here. In our model the regional private providers do not control exclusive meeting places or assortative matching: each prices a distinct local founder population whose realised cross-region partners are complementary. A welfare-maximising public basic-access provider in each region adds an instrument absent from the private matchmaking models; a successful matched venture also competes against incumbent firms in two separate product markets.

Hagiu (2004a,b) study proprietary and open platform design and developers' heterogeneous contributions, while Veiga et al. (2017) emphasise quality-weighted sorting and the differing network values of participants. Our current uniform upper-tail selection obeys *exactly* $q_i = m_i - m_i^2/2$ in every service state, and therefore can be represented by a single nonlinear participation index. We do not claim the innovation of an irreducible second network state, independent mean quality, or the general quality–quantity trade-off. Peitz and Sato (2026) further integrate platform fees, endogenous participation and entry, and Lu et al. (2025) connect network externalities to competitive platform pricing. The specific posted-fee response in our model is a specialisation of familiar pricing and network feedback.

Public alternatives and endogenous private-market responses. Hua et al. (2016) consider free public and paid private services under delay- and capacity-sensitive provision and fiscal policy; the published abstract verifies the presence of this mechanism, but its inaccessible full technical proof does not support a formal theorem-by-theorem exclusion claim. Laine and Ma (2017); Kim (2024) also study mixed public/private market competition. Atal et al. (2024) document the entry of public pharmacies and the resulting segmentation of private customers and increases in private drug prices. The public-private payment-platform model of Liu et al. (2026) demonstrates private merchant-fee increases in the presence of a welfare-oriented public platform, through differentiated preferences and network fragmentation. Therefore public entry, private fee responses, adverse distributional effects and mixed provision are all known. Our public service has no rationing, a private premium does not directly improve matching, and the distinguishing policy object is *regional venture-commercialisation incidence* interacting with a fee-induced basic-access exclusion margin.

Innovation policy, local market incidence and governmental objectives. Takalo et al. (2013) analyse strategic interaction among R&D project applicants, public funding agencies and private financiers with fixed project costs and financing frictions; they demonstrate that subsidy additionality and welfare can have a nontrivial relation. Thus the observation that an innovation-support programme's value cannot be judged by the count of supported projects is not itself new. Our public instrument is *free basic intermediation* rather than project finance or R&D subsidy, and premium payment does not raise the underlying research technology. d'Aspremont and Jacquemin (1988); Goyal and Joshi (2003) study research cooperation and rivalry, while Berliant and Fujita (2008, 2011) address heterogeneous knowledge. Lee (2026)

links local knowledge spillovers and oligopolistic rivalry to sign-dependent R&D intervention. These studies establish that commercialisation benefits need not be positive for every market participant or region. What is specific here is that the sign of local venture-market incidence K_i may be *opposite* to the sign of the derivative of the local government's willingness to restore founder access with respect to technological complementarity.

Long and Wong (2009) model two city governments and two private service firms in inter-city competition; the local-government/private-provider architecture alone is not new. Local and national investment incentives have long differed when cross-jurisdictional public-good benefits are not internalised (Cremer et al., 1997; Mankiw and Whinston, 1986). Accordingly, Lemmas 1–3 are preparatory accounting/selection results, not priority claims. Nor is proof-oriented exact certification of the chosen equilibrium calibration a new theory of competition, although it makes the model's examples more credible.

Economic heterogeneity and policy boundaries. Benassi et al. (2025) revisit canonical mixed-oligopoly models by linking reservation-price distributions to income and asking how distributional changes shift privatization boundaries. Their analysis shows how an established model can answer a different economic question through explicit policy thresholds, while the direction of particular comparisons depends on the specified demand and cost environment. Our heterogeneity concerns founder selection and regional commercialisation incidence, rather than household income or an endogenous privatization share. The relevant comparison is the economic content of the resulting boundary, not the mere presence of additional heterogeneous agents. Here that content is the private exclusion response and the conditional profit-share band; no income-distribution result is imported.

The narrower contribution. The economically distinctive result is Proposition 1, specifically the *primitive profit-share band* (30): in a region where the new venture's commercialisation incidence is negative, higher technical skill complementarity can increase the publicly affordable cost of basic access because profit-related price screening excludes firms from socially valuable direct services. The condition is derived from reoptimised private fees and the local Cournot-derived M_i, not the assumption of a positive local innovation externality. An independent finite-γ interval certificate proves this sign inversion on an explicitly stated continuous primitive rectangle. The model is more structured than a generic nonlinear participation-externality representation, but none of these differences is a formal proof that no general predecessor framework can nest

the result. It is a *conditional, model-specific applied-theory proposition*, not a universal new law of public options.

[Table 3 about here.]

The public-cost interval that supports the three-scenario *ownership switch* remains narrow and the recorded alternative skill distribution removes it. Neither the stronger continuously certified public-value *slope* nor the distinct institutional labels should be used to imply a broad full-game SPNE result, an empirically identified programme effect, or an entitlement to theoretical priority.

8.1 Institutional interpretation and policy boundaries

The British Catapult debate exposes a live tension between public innovation support and private market activity. In March 2026, a House of Commons committee raised concerns that Catapult centres could compete with local organisations for projects and funding, especially where regional R&D activity was weak (House of Commons Science, Innovation and Technology Committee, 2026). The British government's June 2026 response defended mixed public grants, competitively secured research funding and commercial revenues as sustaining market responsiveness and limiting displacement (HM Government, 2026). Neither position is a causal demonstration of crowding out or crowding in. Catapult centres are hybrid organisations and cannot literally be equated with our independent free government entrant G and profit-maximising private entrant P.

The European Digital Innovation Hubs network supplies another motivation. It provides regional access to technical experimentation, expertise and ecosystem connections while enabling cross-region specialist services (European Commission, n.d.). Its operational pricing guidance allows free or discounted assistance alongside market-price-based list values (European Digital Innovation Hubs Network, 2026). Hence zero-priced access and privately priced services may be relevant to the same institutional design; this does not mean every EDIH is state-operated or always free.

A concrete example of capability complementarity is the US Manufacturing Extension Partnership's reported support for a Montana company developing a fibre-optic cable clamp. The publicly supported extension network helped the inventors identify contract-manufacturing capabilities for commercialisation (National Institute of Standards and Technology, 2024). This

illustrates invention-implementation matching but does not estimate an equilibrium treatment effect. Actual extension advice may directly improve technology, whereas our premium service changes direct service value and participation only; that function must not be inferred from the model.

The model's public alternative is a *taxpayer-financed basic service*, not necessarily a physical building. The government finances the real fixed service cost and evaluates regional founder, incumbent and consumer outcomes. Since taxation, building technologies, procurement, subsidies and digital-only matching alternatives are not strategic choices, the study offers diagnostic questions—which skills join, where are new venture profits earned, who loses from incumbent displacement, and how do local incentives diverge from aggregate effects?—rather than a universal construction or ownership recommendation.

9 Conclusion

Private innovation intermediation and public basic access respond to different economic returns. Private fees select founders according to their willingness to pay, which in the maintained model is aligned with technical ability. A government instead compares the cost of access with direct service gains and the regional incidence of additional ventures. The conditional public-cost ceiling connects these decisions to basic and premium service values, consumer and incumbent effects, founder profit shares and cross-region matching.

The strongest analytical result is the primitive profit-share band in Proposition 1. With sufficiently weak matching and full partner participation, stronger skill complementarity can increase the value of restoring basic access even when the resulting ventures reduce regional commercialisation surplus. Positive basic-service value and privately appropriable technical returns jointly sustain this effect. The finite-matching continuous certificate confirms the stated signs within its fixed economic environment. It does not turn the conditional result into an unrestricted equilibrium theorem.

National evaluation adds the other region's welfare increment to the local public-cost ceiling. That known externality principle takes a specific form through privately selected abilities and downstream competition here. The three certified provision examples exhibit both local/national agreement and public underprovision relative to the sixteen-configuration benchmark. Their narrow cost band and the failed distributional and competition-form portability records restrict

their use as policy rules. No general signed market-size or incumbent-count comparative static, optimal corrective policy, or physical-facility recommendation follows.

For evaluation, the model motivates measuring eligible firms, pre-service abilities, private fees, basic and premium use, joint-project outcomes and the regional allocation of consumer, incumbent and venture returns. These are predictions and measurement requirements, not empirical findings. The paper's contribution lies in its conditional regional policy diagnosis and explicit boundaries. Reorganising the results makes that contribution assessable; it neither resolves theorem-level priority against all general predecessors nor establishes that a diagnosis of this scope is sufficient for journal acceptance.

Statements and Declarations

Data and code availability. The accompanying replication materials contain the frozen primitives, all sixteen payoff configurations for each benchmark, rational verification scripts, the diagnostic decomposition and adverse-test records. No new empirical dataset is analysed. Computer-assisted interval checks, analytic proofs and the limited Lean checks have distinct coverage, documented in the Online Supplement; the entire game is not Lean-formalised.

Use of generative tools. Generative AI tools assisted the literature search, mathematical derivations, code, manuscript revision and review checks. Computational checks do not replace author responsibility or external peer review.

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Table 1: Original provider configurations and public-addition thresholds

Technology and λ	Pure NE	Planner	Local threshold
Skills, 0	(B, P)	(B, B)	0.0404250710
Skills, 0.5	(B, B)	(B, B)	0.0405316701
Pooled, 0.5	(B, P)	(B, B)	0.0400432292

B: free basic plus paid premium; P: private only. The last column is region B's public-entry cost threshold conditional on region A remaining in B. The common baseline cost is 0.0405. Planner means the restricted sixteen-configuration welfare benchmark.

Table 2: Region-B public-entry threshold components, with region A fully participating

	Skills, 0	Skills, 0.5	Pooled, 0.5
Direct service	0.0404864338	0.0406350895	0.0402210076
Baseline matching	−0.0000613628	−0.0000614025	−0.0000612921
Technology increment	0	−0.0000420170	−0.0001164863
Total threshold	0.0404250710	0.0405316701	0.0400432292

Entries are gross gains from adding basic access, before the public fixed cost. Baseline matching uses the same $S_B^0 = p_b M_B + \omega_B R_0$ across columns. The pooled technology increment is $C\lambda K_B \ell_L/4$, not the selected $C\lambda K_B \ell_H^2/4$. Display rounding need not sum exactly.

Table 3: Closest prior mechanisms and the conditional result retained here

Closest prior research	Existing mechanism and distinct policy object
Damiano and Li (2007, 2008); Hagiu (2004b)	Complementary-type matching, screening through admission prices and quantity–quality trade-offs are known; we value the restoration of fee-excluded basic services using local post-innovation market surplus.
Atal et al. (2024); Liu et al. (2026); Hua et al. (2016)	Public provision, private fee/participation responses and mixed services are known; our policy-value derivative varies with R&D skill complementarity and locally adverse commercialisation.
Takalo et al. (2013); Lee (2026)	Innovation subsidies, adverse incumbent effects and sign-sensitive externalities are known; our government chooses zero-priced access against a private posted-fee selected population, not an R&D grant.
Peitz and Sato (2026); Cremer et al. (1997)	Endogenous network participation and regional government wedges are established; our contribution is a primitive-level, conditional sign conflict with an exact finite-γ continuum witness.